Week 10

Security Threats and Robust Federated Learning

Poisoning and robust aggregation

Graduate course · English instruction



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Privacy and security in FL

Privacy

Can an update or the final model reveal private traini ng information?

Example: recovering details of a client’s data.

Security and robustness

Can a faulty or malicious client damage the model?

Example: changing predictions by sending a harmful update.

Learning objectives

Attack types

Distinguish changes to data from changes to transmitted updates.

Aggregation

Explain the mean, median, trimmed mean, and Krum.

Assumptions

Explain why honest non-IID updates can look unusual.

Evaluation

Compare clean and attacked runs with a clear threat model.

Threats describe different parts of an attack

Data poisoning

Changes local examples or label s.

Model poisoning

Directly changes the update sent to the server.

Backdoor

A targeted goal: trigger inputs pro duce an attacker-chosen label.

Byzantine behavior

Arbitrary behavior caused by fault s or malicious clients.

Collusion

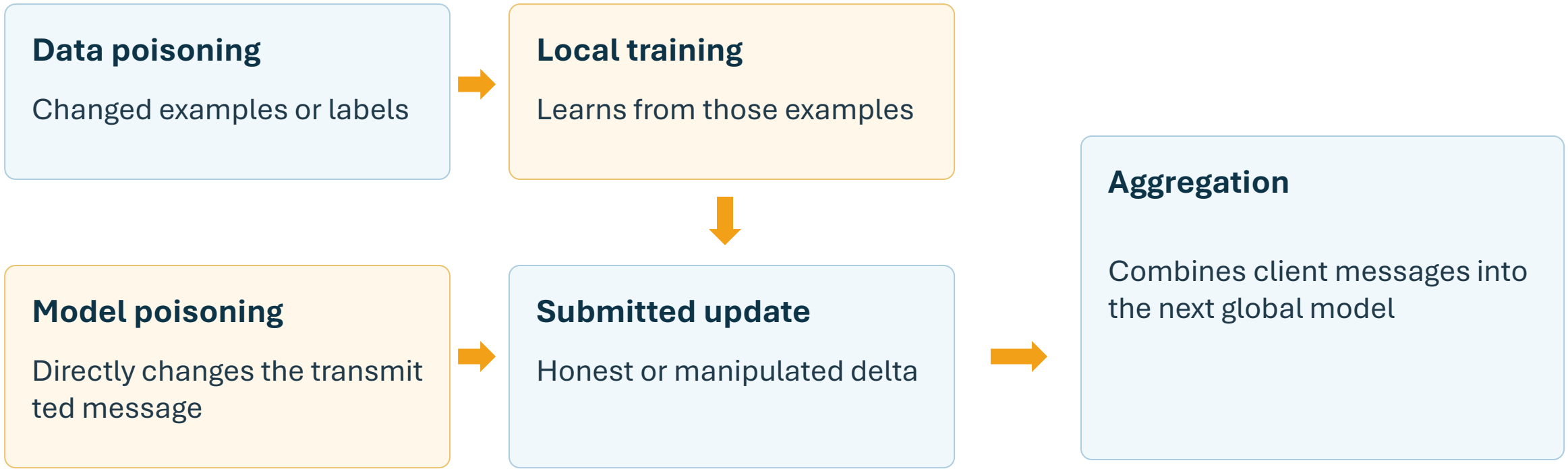
Several malicious clients coordin ate.

Sybil participation

One attacker controls several ap parent client identities.

Method reference: Kairouz et al. (2021); Bagdasaryan et al. (2020)

Where poisoning enters training



Two entry points can affect the same aggregation step.

Method reference: Bagdasaryan et al. (2020)

Backdoors need a separate test

Clean inputs

The model may still predict normal examples correctl y.

Triggered inputs

An added pattern causes an attacker-chosen predicti on.

Clean accuracy

Measure correct predictions on untouched test data.

Attack success rate

Measure target predictions on eligible triggered input s. State the denominator.

Method reference: Bagdasaryan et al. (2020)

Non-IID data makes detection harder

Honest variation

A client with rare labels can send a useful but unusual update.

False alarm

Rejecting that update may harm an underrepresented group.

Missed attack

A harmful update can look similar to ordinary updates.

Evaluation need

Report the data split and performance across classes or clients.

Ludwig & Baracaldo (2022), Chs. 16–17 · Kairouz et al. (2021), §5

One extreme value can move the mean

Seven values for one model coordinate

0.8	0.9	1.0	1.1	1.2	1.3	9.0

Mean = 2.19

15.3 ÷ 7
Sensitive to the extreme value.

Median = 1.10

Middle of the sorted values.

Trimmed mean = 1.10

Remove 0.8 and 9.0, then averag e.

Trimming removes extreme values, including honest ones.

Method reference: Teaching example; Yin et al. (2018)

Coordinate-wise median

For each coordinate j, take the median across client updates.

$$a_j = \text{median}(\Delta_{1,j}, \ldots, \Delta_{n,j})$$

Each coordinate uses its own ordering. The result may differ from every submitted vector.

A strict majority of honest clients limits arbitrary scalar outliers. Learning accuracy also depends on how honest updates vary.

Method reference: Yin et al. (2018)

Trimmed mean

Sort the n values for each coordinate. Remove b values at each end.

a_j = mean of sorted values b + 1 through n − b

Example: n = 20 and b = 4 retain 12 values per coordinate.

Need 2b < n. Choose b with an attack-budget bound in mind, then check clean-data bias and non-IID behavior.

Method reference: Yin et al. (2018)

Krum selects a whole update

Distance score

For each update, sum squared distances to its n − f − 2 nearest other updates.

Selection

Choose the update with the smallest score. Here f bo unds Byzantine clients.

Original conditions

n > 2f + 2, plus assumptions on honest gradient estim ates and their noise.

FL limitation

Different local data and multiple local steps can wea ken those assumptions.

Method reference: Blanchard et al. (2017)

What each defense can and cannot do

Method	What it controls	Main limitation
Median	Extreme values per coordinate	Honest client variation can bias the result.
Trimmed mean	Tails of each coordinate’s distribution	Needs a trimming level and enough retained values.
Krum	Selection using distances between vectors	Count and statistical conditions must hold.
Clipping	The norm of each individual update	A small harmful update can still pass.

A defense needs both a threat model and evidence under that model.

Method reference: Yin et al. (2018); Blanchard et al. (2017)

Defense beyond aggregation

Trusted validation

Check candidate models on representative trusted d ata, if available.

History and identity

Track behavior over time and limit uncontrolled client identities.

Trusted execution

Verify approved code where hardware and trust assu mptions allow it.

Limits remain

Good history, approved code, or personalization alon e does not establish robustness.

Ludwig & Baracaldo (2022), Chs. 16–17 · Kairouz et al. (2021), §5

Update privacy changes available defenses

Visible individual updates

The server can calculate norms and distances or insp ect each contribution.

This visibility can expose information about clients.

Secure sums

Standard secure aggregation reveals a sum while hidi ng individual contributions.

A sum alone is insufficient for ordinary median, Krum , or per-client inspection.

Ludwig & Baracaldo (2022), Chs. 16–17 · Kairouz et al. (2021), §5

A clear evaluation starts with the threat model

Attacker control

How many selected clients are malicious? Can they c hange data, updates, or identities?

Attacker knowledge

Does the attacker know the defense or observe hone st updates?

Training setting

State non-IID partitioning, participation, local steps, and weighting.

Evidence

Report clean and attacked results, multiple seeds, an d the right attack metric.

In-class attack and defense mapping

Case 1

A client deliberately changes many labels at random.

Case 2

Five clients coordinate updates to create a trigger-ba sed error.

Case 3

A slow device submits an update based on an old glo bal model.

Discussion

Identify the control and goal. Propose a defense and state one limit.

Colab lab setup

Task and model

8 × 8 digit classification with softmax regression. Publ ic scikit-learn digits data.

Clients and attack

20 simulated clients, all active. Four send −8 times th eir own update.

Controlled comparison

Mean, median, and trimmed mean. Clean and attack ed runs, IID and non-IID data.

Reproducibility

50 rounds, 3 local steps, learning rate 0.3. Seeds 11, 22, 33. CPU runtime.

From local training to a global update

```
delta = local_update(W, X_client, y_client)

if malicious:
    delta = -8.0 * delta

step = aggregate(updates, rule)
W = W + step
```

Meaning of delta

Local model minus global model. The s erver adds the aggregated change.

Fair comparison

Same split, clients, initial weights, and training settings for each rule.

Full functions, explanations, plots, and checks are in the notebook.

What the experiment supports

Observed

Coordinate-wise rules reduce damage from this large sign-flip attack.

Trade-off

Non-IID data lowers clean and attacked accuracy for these defenses.

Not established

No guarantee against adaptive attacks, backdoors, or other client distributions.

Next experiment

Vary one factor at a time, use validation for tuning, an d report more seeds.

Key takeaways

1. An attacker can change local data or directly change an update.
2. Backdoors require triggered-input testing as well as clean accuracy.
3. Robust aggregation depends on client and data assumptions.
4. Honest non-IID updates can be useful outliers.
5. Privacy protocols change which defenses the server can run.

Reading and preparation

Core methods	Yin et al. (2018), Byzantine-Robust Distributed Learning: Towards Optimal Statistical Rates. I CML.
Whole-vector selection	Blanchard et al. (2017), Machine Learning with Adversaries: Byzantine Tolerant Gradient Des cent. NeurIPS.
Targeted poisoning	Bagdasaryan et al. (2020), How To Backdoor Federated Learning. AISTATS.
Course background	Kairouz et al. (2021), Advances and Open Problems in Federated Learning. Ludwig and Bara caldo, eds. (2022), Federated Learning.

Q & A

Thank You.

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